



Immersive Three-Dimensional Modeling and Virtual Reality for Enhanced Visualization of Operative Neurosurgical Anatomy

Samuel B. Tomlinson^{1,2}, Benjamin K. Hendricks^{1,3}, Aaron Cohen-Gadol^{1,4}

Learning the endless intricacies of operative neurosurgical anatomy requires that surgeons complement their intraoperative experiences with a variety of educational resources. In the past 2 decades, rapid improvements in digital graphics and computing power have enabled a new generation of 3-dimensional (3D) virtual resources that overcome limitations of more traditional 2-dimensional materials. Today, dozens of immersive 3D visualization platforms exist for applications such as learning neuroanatomy, simulating operative techniques, and planning surgical interventions with patient-specific models. The purpose of this article is to identify current applications of 3D digital modeling and virtual reality in neurosurgery. In addition, we showcase a new series of freely available 3D virtual-reality models created to assist in learning complex cranial anatomy. We anticipate these models to have a wide range of educational, clinical, and research applications. Three-dimensional visualization is poised to modernize the ways we learn and teach neurosurgical anatomy outside of the operating room. Future generations of neurosurgeons are expected to benefit from these technologies from the earliest stages of training.

INTRODUCTION

Mastering the intricacies of cranial and spinal anatomy is a lifelong endeavor at the heart of every neurosurgeon's practice. For generations, neurosurgeons have

supplemented their intraoperative learning with a host of traditional resources, including cadaveric dissection, preserved specimens, and various 2-dimensional (2D) image representations (e.g., textbook illustrations, autopsy photographs, atlases, tomographic scans).¹ Surgeons operating within the delicate corridors of the central nervous system require a comprehensive understanding of 3-dimensional (3D) anatomy and complex spatial relationships. Whereas cadaveric dissection and other resource-intensive physical models provide valuable experience with 3D reasoning, 2D learning resources suffer inherent limitations.²

Twenty-first century innovations in computing power, 3D digital design, and virtual reality have revolutionized the pedagogical toolkit for learners of surgical anatomy.^{3,4} Today, neurosurgeons at all training levels can access detailed, interactive, Web-based 3D virtual models of neuroanatomical systems from their personal computing devices, often completely free of charge.⁵⁻⁹ Many packages enable representation of patient-specific imaging data to facilitate preoperative case planning, trainee education, and surgeon-patient conversations.¹⁰⁻¹² By layering additional technologies atop these platforms, such as virtual manipulation and haptic feedback, surgeons can hone their operative technique in risk-free and increasingly realistic simulation environments outside of the operating room.^{13,14} The creation of patient-specific 3D-printed physical models is another exciting elaboration of this technology that has rapidly expanding applications in neurosurgery.¹⁵

Our group recently introduced a series of freely accessible, interactive, and ultrarealistic online 3D virtual models designed to facilitate learning of intricate cranial and cerebrovascular anatomy.^{5,7,16} The goal of this article is to contextualize our development within the broader movement to incorporate virtual technologies into modern neurosurgical education.

Key words

- 3D
- *The Neurosurgical Atlas*
- Operative anatomy
- Resident education
- Virtual reality

Abbreviations and Acronyms

2D: 2-dimensional

3D: 3-dimensional

PSP: Pedicle screw placement

VHP: Visible Human Project

From the ¹Neurosurgical Atlas, Indianapolis, Indiana; ²School of Medicine and Dentistry, University of Rochester Medical Center, Rochester, New York; ³Department of Neurosurgery, Barrow Neurological Institute, Phoenix, Arizona; and ⁴Goodman Campbell Brain and Spine and Department of Neurological Surgery, Indiana University School of Medicine, Indianapolis, Indiana, USA

To whom correspondence should be addressed: Aaron Cohen-Gadol, M.D., M.Sc., M.B.A. [E-mail: acohenmd@gmail.com]

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VIRTUAL TECHNOLOGIES IN NEUROSURGERY EDUCATION

In this article, “virtual anatomy”¹⁷ refers to the general process of using computers to create digital models that allow users to visualize neuroanatomy in 3 dimensions. Efforts to build a comprehensive 3D virtual model of the human brain have a long history.² Much credit for early innovation is owed to the visionaries behind the Visible Human Project (VHP), who in the 1990s used funding from the National Library of Medicine to create an electronic database of cross-sectional and 3D human anatomy from 2 anatomical donors (“visible male” and “visible female”).^{18,19} Educational resources built on freely licensed VHP data can be found in both the commercial and academic sectors,²⁰ including a wave of 3D virtual neuroanatomy atlases (eg, Voxel-Man 3D-Navigator: Brain and Skull^{21,22}). Concurrent initiatives were launched at the institutional level, and notable advances in digital brain and spine modeling have emerged from the University of Washington’s Digital Anatomist program.^{23–25}

The prolific work of Wieslaw L. Nowinski and colleagues also should be celebrated; for decades, Nowinski et al. have defined the state-of-the-art for 3D digital brain atlases covering normal and disordered central nervous system states,^{6,8,9,26–39} which have been distributed worldwide through academic and commercial licenses. The group’s most ambitious effort to date—to construct “an advanced, detailed, parcellated, labeled, accurate, interactive, 3-dimensional, and scalable whole human brain atlas”⁸—has been realized in the form of a freely accessible digital atlas that incorporates thousands of 3D structures of the head and neck,⁹ with continued improvements undoubtedly forthcoming.

Although holistic virtual 3D brain models provide unparalleled breadth, detailed regional models have been developed to assist with learning the specific intricacies of the temporal bone,^{16,40–42} clivus,⁴³ cavernous sinus,⁴⁴ skull base,^{29,45,46} middle ear,⁴⁷ and, recently, the entire cranium.⁵ A 3D virtual model of the temporal lobe was created by de Ribaupierre and Wilson⁴⁸ for teaching the surgical anatomy of temporal lobectomy, and other groups have created 3D representations of the cortical surface and superficial vasculature to assist with surgical approach planning.^{49,50} At the systems level, 3D atlases exist for studying cranial nerve anatomy,^{32,36,37,51,52} white matter pathways,³³ and the cerebrovascular system,^{27,30,35,39,53} including our own comprehensive vasculature models.⁵⁴ These models of normal anatomy have served as the foundation for disease-oriented models, such as an innovative 3D stroke atlas that correlates vascular pathology with neuroanatomical and clinical sequelae.³⁸

Results of the few studies comparing the educational value of 3D virtual-anatomy resources against conventional methods have generally been mixed,^{51,55–59} although some evidence in the neurosurgical literature is positive.¹⁶ Brewer et al.⁶⁰ used manual segmentation of the VHP Visible Male data set to generate a 3D virtual brain model targeted at early-stage medical students. Whereas students who completed self-study with the 3D model performed similarly to students in the conventional 2D group on an anatomy post-test, most students preferred learning with the 3D model. Similar results were found in a randomized controlled study by Stepan et al.,⁶¹ who assigned 66 medical students to receive either 3D virtual-reality self-study materials or 2D online textbook images. Performances on an anatomy post-test were

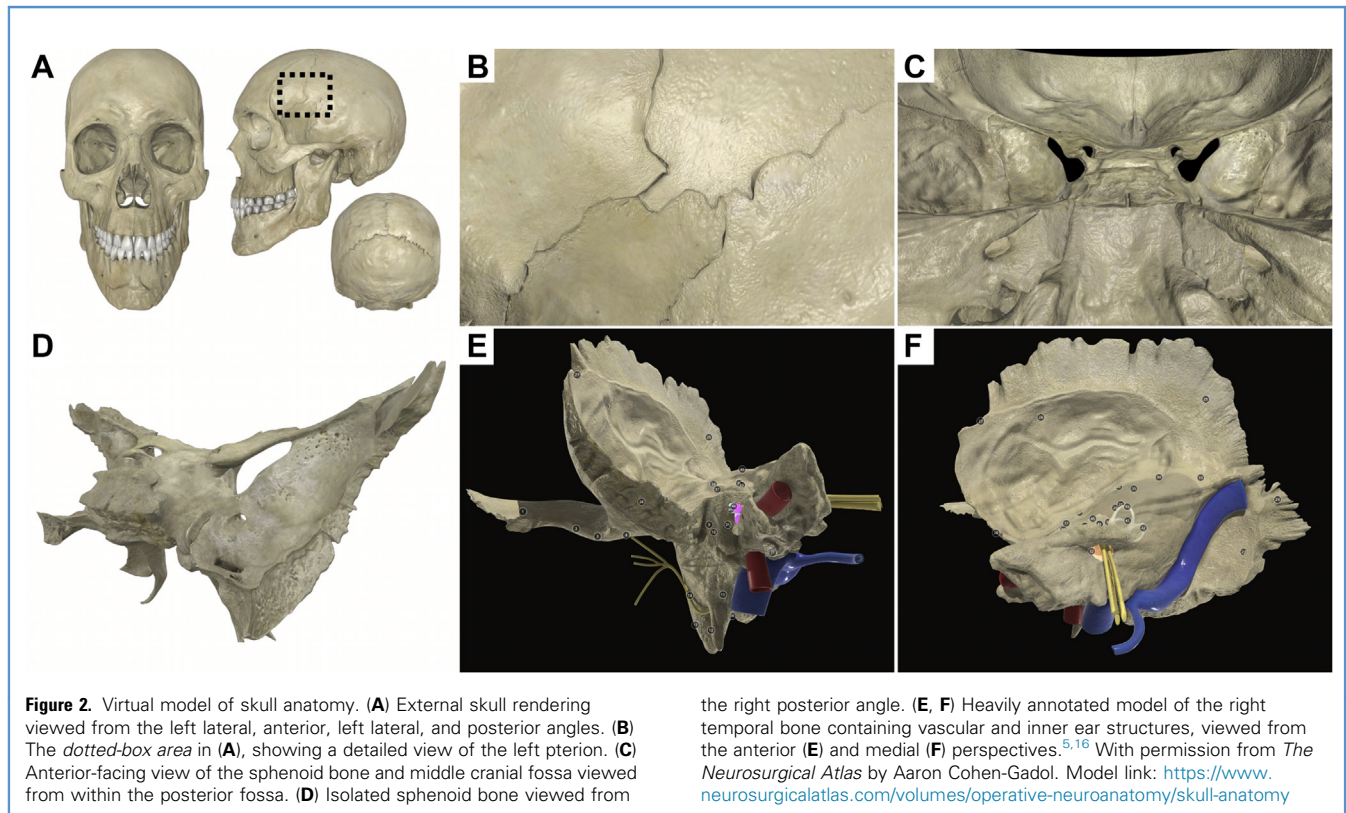
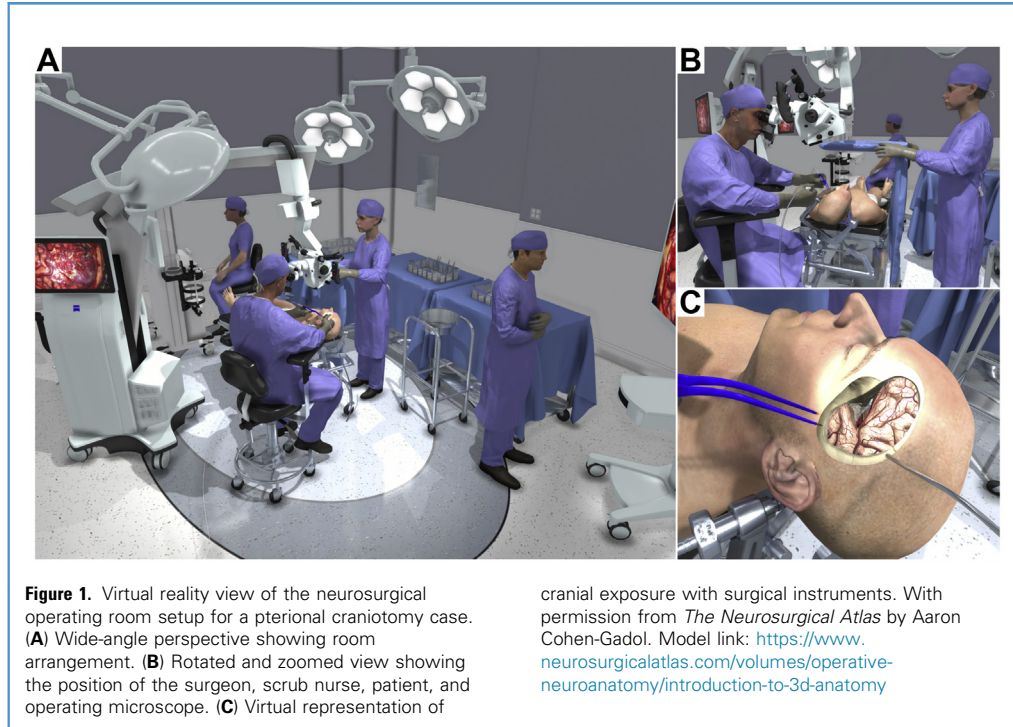
similar between the experimental groups, but the 3D group rated the experience as more engaging and scored higher on a learner motivation survey.

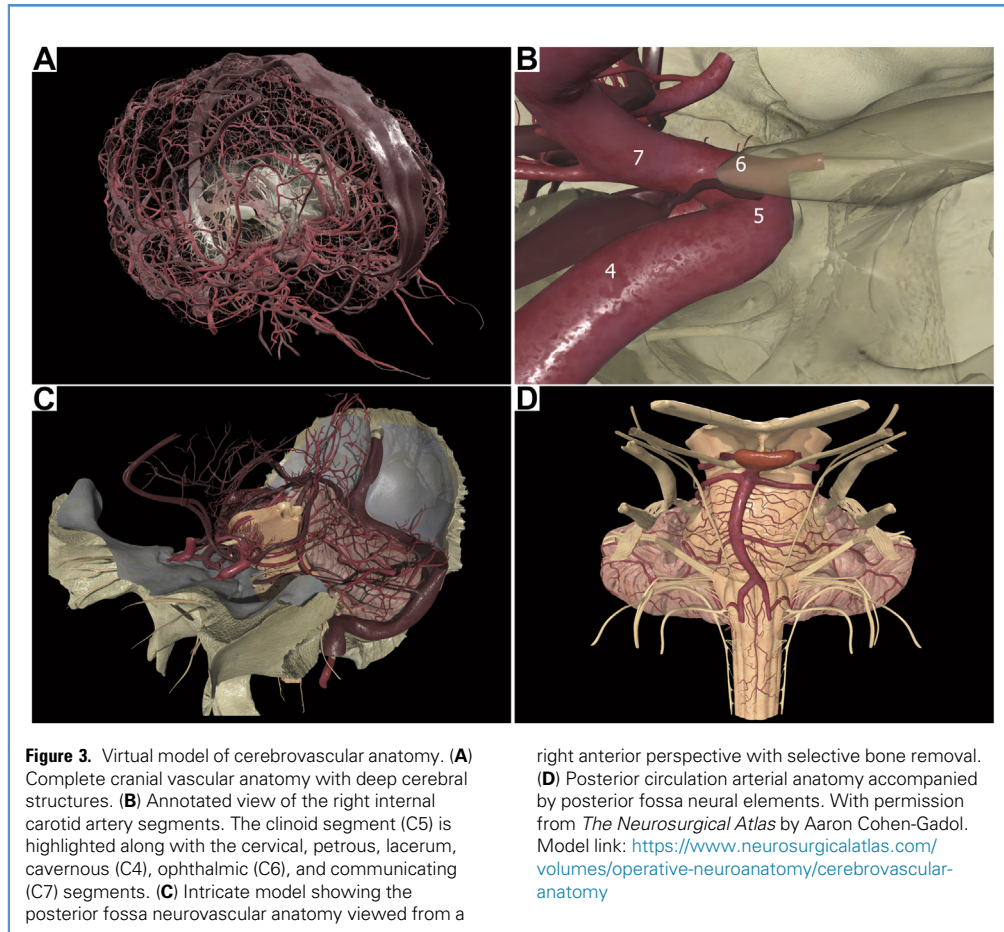
In a rare quantitative demonstration, Ruisoto et al.⁶² assigned medical students to study either 3D volumetric or 2D cross-sectional images before an anatomy post-test. Participants in the 3D group scored greater on accuracy and on measures of response time, confidence, and enthusiasm. Similar quantitative benefits of 3D digital neuroanatomical models on learning were reported in a cross-over study by Allen et al.⁶³ Interestingly, several studies have found that baseline visuospatial abilities correlate with the effectiveness of 3D learning exercises,^{60,64,65} suggesting that certain individuals benefit more or less from these resources.

“Virtual surgery” is an even more recent expansion of 3D digital technologies allowing users to manipulate virtual objects in a fashion that mimics that of actual surgery.¹² Diverse applications of neurosurgical simulation have emerged in the past 2 decades as intensifying regulatory pressures continue to reshape traditional apprenticeship models of surgical education.⁶⁶ For students and residents, simulation can expedite the acquisition of fundamental skills and identify areas of weakness outside the operating room without compromising patient safety.^{10,67} Advanced surgeons might benefit from immersive visualization of complex anatomy for case planning. Object manipulation in these 3D environments is designed to incorporate intuitive hand movements, and most applications combine patient-specific 3D renderings with haptic feedback and tissue deformation for maximal realism.^{10,68}

Virtual surgery applications are extremely varied but typically aim to facilitate neuroanatomical learning, surgical planning, and/or procedural simulation.⁶⁸ Within the lattermost category, the earliest contributions came in the form of simulators for ventriculostomy,⁶⁹ an essential procedure learned in the beginning stages of residency. Several ventriculostomy simulators, including the ImmersiveTouch platform (ImmersiveTouch, Inc., Chicago, Illinois, USA), have been developed, and training in this virtual environment is suggested to approximate the target accuracy and learning curve of real-world ventriculostomy training.^{70–72} Other procedural neurosurgery simulators emerged more recently for microvascular decompression,⁷³ aneurysm clipping,^{74–78} tumor surgery,^{79–81} temporal bone dissection,^{82,83} endoscopic techniques,⁸⁴ and pedicle screw placement⁸⁵ and for learning the various surgical corridors.^{43,86} Complementing virtual reality with 3D-printed physical models (i.e., “mixed reality”) also has been used to produce spectacularly realistic models, such as a recent mixed-reality simulator for endoscopic third ventriculostomy in children.⁸⁷ At several institutions, virtual- and mixed-reality simulators have been incorporated into the routine presurgical workflow to assist in case planning^{88–90} or into the formal residency training curriculum.⁶⁷

As with other ascendant technologies, innovation has outpaced evidence proving its beneficial effect on patient-centered outcomes such as complication rates and recovery time. However, some evidence in the neurosurgery literature points to an educational benefit for simulation-based learning over more conventional approaches. For example, in an interesting study that examined pedicle screw





placement (PSP), Xin et al.⁸⁵ randomly assigned 16 resident-level surgeons inexperienced with PSP to practice with a virtual-reality simulator (experimental) or to study a 3D-printed spinal model (control). Each participant then performed bilateral T-11 to L-4 PSP on a cadaver. Both the accuracy and efficiency of screw placement were superior in the virtual-reality–trained group than in the control group. Unfortunately, evidence of this kind remains sparse in the literature,⁵⁶ and perceptions of the cost–benefit trade-off for these expensive technologies are likely to vary based on institutional goals and resources.

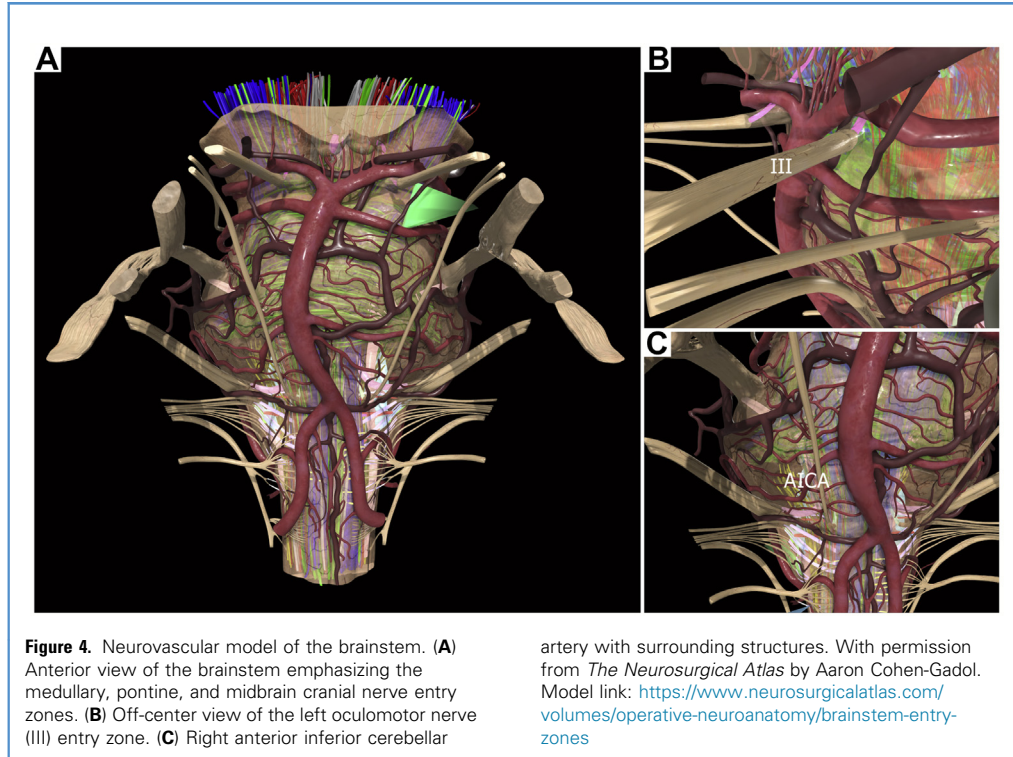
CREATION OF 3D VIRTUAL MODELS: THE NEUROSURGICAL ATLAS EXPERIENCE

Introduced in 2016, *The Neurosurgical Atlas* (www.neurosurgicalatlas.com) is a free, worldwide educational resource that publishes multimedia content related to micro-neurosurgery. The popularity of the website has increased steadily over the past several years, and its majority demographic (~45% of visitors are between 25 and 34 years old⁹⁷) includes medical students, residents, and early-career neurosurgeons poised to benefit from effective learning resources.

Through the combined efforts of neurosurgeons, anatomists, and 3D computer graphics artists, *The Neurosurgical Atlas* has

released a series of interactive 3D virtual-reality models that provide a tour through the neurosurgical operating room (Figure 1) as well as many critical aspects of cranial surgical anatomy (Figures 2–5).^{5,7,16} The anatomical models were constructed from a combination of computed tomography scans, catheter angiograms, and 1.5-T 3D magnetic resonance imaging. Imaging data collected from patients were processed using Amira (Thermo Fisher Scientific, Waltham, Massachusetts, USA) and Maya (Autodesk, San Rafael, California, USA) to generate a 3D based polygonal mesh ready for sculpting, texturizing, and hand-painting via graphic design software (details provided elsewhere^{5,16}). The 3D models can be displayed on a mobile phone, computer, or a virtual-reality headset to provide a more immersive experience. Individual model elements can be selected and hidden to optimize viewing of desired structures. The models do not currently incorporate tissue deformation or haptic feedback for procedural simulation, although these are exciting future possibilities.

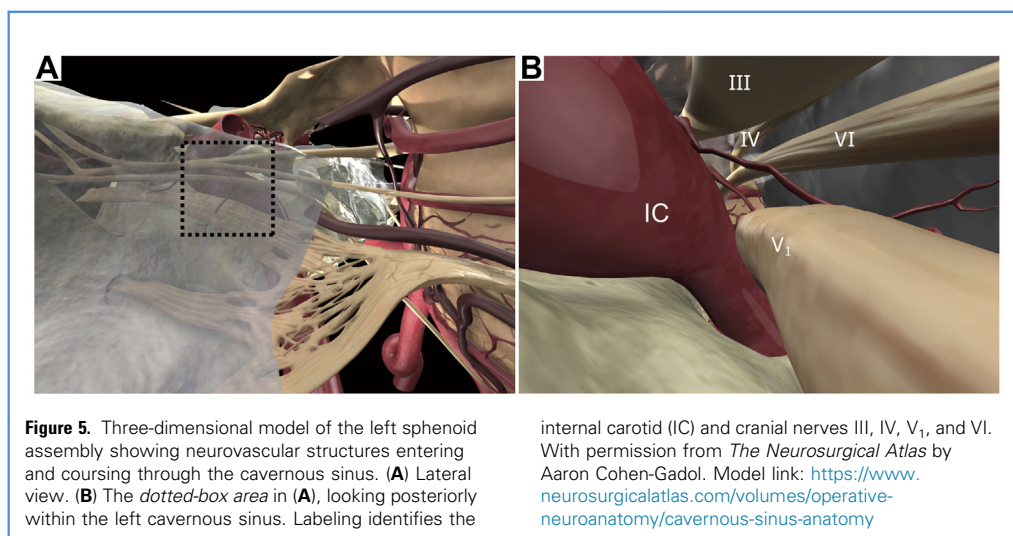
The virtual anatomy models currently encompass skull anatomy⁵ (Figure 2), the cerebrovascular system⁵⁴ (Figure 3), brainstem anatomy (Figure 4), and the cavernous sinus (Figure 5). Each model is parcellated and color-coded to distinguish regional neuroanatomy, and virtual objects can be freely rotated, zoomed, and navigated by toggling between dozens of built-in annotations.



Early survey data obtained from neurosurgical residents who viewed the temporal bone model revealed that >90% of residents preferred to study with the 3D resource versus conventional 2D textbook illustrations.¹⁶ We anticipate that these detailed interactive models will prove tremendously valuable for users of all experience levels, and studies evaluating their quantitative learning efficacy are currently underway.

SUMMARY AND CONCLUSIONS

Innovations in 3D digital technology have enabled us to create educational resources that are vastly more sophisticated and realistic than traditional 2D materials. Future generations of neurosurgeons will likely benefit from these technologies from the earliest stages of their training, beginning with 3D atlases of normal neuroanatomy and gradually incorporating patient-specific



visualization or virtual-reality simulation as their needs become more advanced. In this article, we have attempted to portray the scope of 3D digital technologies in modern neurosurgery education. We have also highlighted our group's recent contribution—a series of immersive 3D virtual models of the skull and cranial contents—with the hope that these free online resources will help

learners of all experience levels tackle the endless visuospatial complexities of neurological surgery.

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